

随机过程作业七参考答案

习题五 19, 22, 25

19、设平稳过程 $\{X(t); -\infty < t < \infty\}$ 的谱密度为 $S_X(\omega) = \frac{1}{\omega^4 + 5\omega^2 + 6}$, 求 $\{X(t)\}$ 的自相关函数

解

$$S_X(\omega) = \frac{1}{\omega^4 + 5\omega^2 + 6} = \frac{1}{\omega^2 + 2} - \frac{1}{\omega^2 + 3}$$
$$\therefore R_X\tau = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{1}{w^2 + 2} - \frac{1}{w^2 + 3} \right) e^{iw\tau} dw = \frac{\sqrt{2}}{4} e^{-\sqrt{2}|\tau|} - \frac{\sqrt{3}}{6} e^{-\sqrt{3}|\tau|}$$

22、已知平稳过程 $\{X(t); -\infty < t < \infty\}$ 的谱密度如下:

$$S_X(\omega) = \begin{cases} 2\delta(\omega) + 1 - |\omega|, & |\omega| < 1 \\ 0, & \text{其他} \end{cases}$$

求 $\{X(t)\}$ 的自相关函数

解

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(w) e^{iw\tau} dw$$
$$= \frac{1}{2\pi} \left(\int_{-1}^1 2\delta(w) e^{iw\tau} dw + \int_{-1}^1 e^{iw\tau} dw - \int_{-1}^0 (-w) e^{iw\tau} dw - \int_0^1 2we^{iw\tau} dw \right)$$
$$= \frac{1}{\pi} \left[1 + \frac{2\sin^2(\frac{\tau}{2})}{\tau^2} \right]$$

25、设平稳过程 $\{X(t); -\infty < t < \infty\}$ 的谱密度为 $S_X(\omega)$, 令 $Y(t) = X(t+L) - X(t)$, 证明: $\{Y(t)\}$ 的谱密度为

$$S_Y(\omega) = 2S_X(\omega)(1 - \cos \omega L).$$

解

$$R_Y(\tau) = E[Y(t)Y(t+\tau)] = E\{[X(t+L) - X(t)][X(t+\tau+L) - X(t+\tau)]\}$$

展开并利用 $X(t)$ 平稳性:

$$R_Y(\tau) = 2R_X(\tau) - R_X(\tau-L) - R_X(\tau+L)$$
$$S_Y(\omega) = \int_{-\infty}^{\infty} R_Y(\tau) e^{-i\omega\tau} d\tau$$
$$= 2S_X(\omega) - e^{-i\omega L} S_X(\omega) - e^{i\omega L} S_X(\omega)$$
$$= S_X(\omega) [2 - (e^{i\omega L} + e^{-i\omega L})]$$
$$= S_X(\omega) (2 - 2\cos \omega L) = 2S_X(\omega) (1 - \cos \omega L)$$